

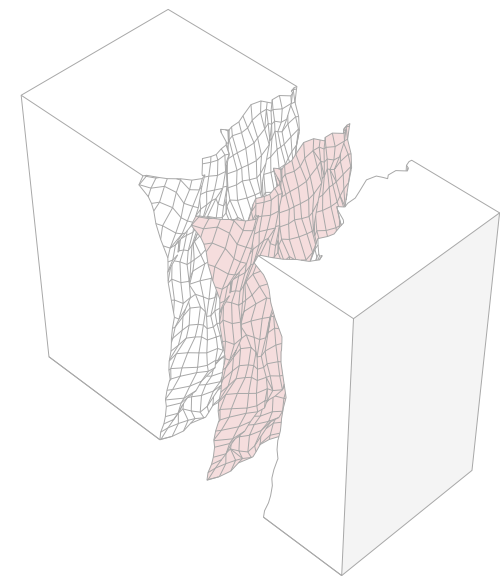


libharmonics : A C++ Library for Morphological Analysis and Spectral Galerkin Methods

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Introduction

A modern C++ library for morphological analysis and spectral Galerkin methods on parametric surfaces.

Motivation

This theme faithfully reproduces the visual identity of the original EPFL LaTeX Beamer class. All colours, spacing, and layout elements are matched.

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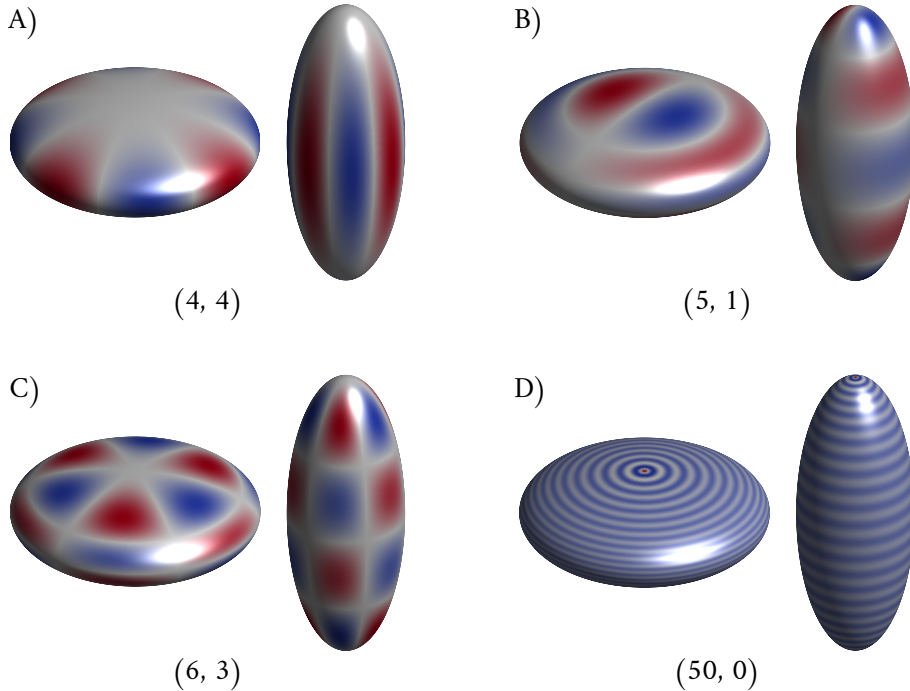
Why Typst?

Typst offers a modern alternative to LaTeX for academic slides and documents:

1. Fast compilation
2. Clean, readable markup
3. Native Unicode support
4. No dependency headaches

Methodology

Spectral parameterization



$$\begin{pmatrix} x(\eta, \varphi) \\ y(\eta, \varphi) \\ z(\eta, \varphi) \end{pmatrix} = \sum_{n=0}^{n_{\max}} \sum_{m=-n}^n A_n^m S_n^m(\eta, \varphi).$$

This spectral expansion maps the surface coordinates (x, y, z) onto a basis of spherical harmonics S_n^m , weighted by coefficients A_n^m .

Figure 1: Oblate and prolate spheroidal harmonics basis functions.

Step-by-step Reveal

1. First point is always visible
- 2.
- 3.

Step-by-step Reveal

1. First point is always visible
2. Second point appears on click
- 3.

Step-by-step Reveal

1. First point is always visible
2. Second point appears on click
3. Third point appears on next click

Note: This conclusion only appears with the third point.

```
+ First point is always visible
#pause
+ Second point appears on click
#pause
+ Third point appears on next click

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Pillar One: Theoretical Framework

The theoretical framework extends established results to handle domain-specific complexities.

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Pillar Two: Computational Methods

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Key insight: computation scales linearly with input size.

Results

Discussion

- **+25 %** accuracy improvement
- **3×** faster than prior art
- Robust across varied conditions

Conclusion

Conclusion

Conclusion

This work presents a comprehensive framework combining theoretical rigour with practical efficiency. Future work will explore extensions to higher-dimensional problems and real-time applications.

References

- [1] M. Shaqfa and W. M. van Rees, "Spheroidal harmonics for generalizing the morphological decomposition of closed parametric surfaces," *Construction and Building Materials*, vol. 454, p. 138967, Dec. 2024, doi: 10.1016/j.conbuildmat.2024.138967.
- [2] M. Shaqfa, "Morphology-preserving remeshing approach to particulate microstructures via harmonic decomposition," *Powder Technology*, vol. 470, p. 121991, Mar. 2026, doi: 10.1016/j.powtec.2025.121991.

Thank You

Questions?

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